Navigation QA report: SP2613_169851

MPL Internship Program Day at Sea

PI Pereira, Filipe · vessel Robert Gordon Sproul · 2026-07-09 to 2026-07-09 · expected area 32.5997°N to 32.7054°N, 117.377°W to 117.226°W · contact Rolling Deck to Repository (R2R)

Source: data/SP2613_169851_gnss · 1 of 2 nav files · format nav29 (STARC logger: timestamp, then NMEA)

Receiver: com.trimble BX992; talker IDs: GN (multi-constellation GNSS, 56925 fixes)

Generated 2026-10-29 10:18 UTC by nav-qa-processing 0.1.0, numpy 2.5.3, pandas 3.0.6, pyproj 3.8.0, shapely 2.2.0

All 8 checks pass.

Warning: 1 of 2 nav files are outside the cruise window (2026-07-09 to 2026-07-09), with fixes from 2026-07-07 20:25 to 2026-07-07 20:25 UTC: SP2613_mru_trimble_bx992-2026-07-07.txt.

R2R fileset 169851 · vessel ID 32QU · San Diego to San Diego · receiver com.trimble BX992 · R2R format 100150 (starc_raw)

R2R rating: GREEN (G)

GREEN (G) if all tests GREEN, RED (R) if at least one test RED, else YELLOW (Y)

R2R tests

Test	Rating	Result	G	Y	Detail
PercentCompleteness Percentage of sampling intervals from the first fix to the last that hold a fix.	G	99.98 percent	≥ 95.00	≥ 75.00	56925 of 56938 intervals of 0.5 s
LongestGapEpoch Longest time between successive fixes.	G	7.000 s	≤ 900.000	≤ 86400.000	0 gaps of the gap threshold or longer
PercentRecordsOutOfSequence Percentage of fixes timed before the fix preceding them.	G	0.00 percent	≤ 5.00	≤ 10.00	0 of 56925
PercentRecordsWithBadQualityIndicator Percentage of fixes whose GGA quality indicator is not a valid fix.	G	0.00 percent	≤ 5.00	≤ 10.00	0 of 56925 reporting one
PercentUnreasonableSpeeds Percentage of speeds between successive fixes above the speed threshold.	G	0.00 percent	≤ 5.00	≤ 10.00	0 of 56924
PercentUnreasonableAccelerations Percentage of fixes whose acceleration is above the acceleration threshold.	G	0.00 percent	≤ 5.00	≤ 10.00	0 of 56917

Infos

Info	Value
TotalTests	6
GTests	6
YTests	0
RTests	0

Ranges

Over every record before QC, as R2R reports them; the statistics below are over the fixes that pass QC.

Range	Quantity	Minimum	Maximum
NSV	Number of satellites used in the fix	10	14
HDOP	Horizontal dilution of precision	0.7	1.0
Altitude	Antenna height above mean sea level [m]	5.494	10.425
HorizontalSpeed	Speed over ground from successive positions [m s-1]	0.00	4.81
HorizontalAcceleration	Horizontal acceleration [m s-2]	-0.455	0.519

Checks

Check	Status	Value	Detail
Completeness	pass	99.98 %	56925 of 56938 0.5 s epochs
Longest gap	pass	00:00:07.000 (7.000 s)	0 gaps of 300 s or longer
Records out of sequence	pass	0.00 %	0 records
Bad quality indicator	pass	0.00 %	0 of 56925 records
Unreasonable speeds	pass	0.00 %	0 of 56924 above 8.7 m s-1
Unreasonable accelerations	pass	0.00 %	0 of 56917 above 1 m s-2
Within the expected area	pass	0 outside	32.5997°N to 32.7054°N, 117.377°W to 117.226°W
Over water	pass	0 over land	more than 10000 m from Natural Earth's 1:10m coasts and rivers
Fixes flagged by QC	info	0	

Checks by file

What each check counted, per nav file, and the sentences the reader couldn't use (the reader and manifest errors say why). A speed, a step out of sequence or a gap is counted in the file of the fix that ends it.

File	Number of fixes	Skipped sentences	Out of sequence	Bad quality	Unreasonable speeds	Unreasonable accelerations	Gaps	Outside area	Over land	Flagged by QC
SP2613_mru_trimble_bx992-2026-07-09.txt	56925	0	0	0	0	0	0	0	0	0
SP2613_mru_trimble_bx992-2026-07-07.txt	skipped: outside the cruise window									
Total	**56925**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

Statistics

Over the fixes that pass QC, but for the times between fixes, which are over every usable record, as the checks count them.

Quantity	Minimum or value	Maximum
Number of fixes that pass QC	56925	
First and last fix	2026-07-09T15:11:57.50Z	2026-07-09T23:06:26.00Z
Most common time between fixes [s]	0.500	
Longest time between successive fixes [hh:mm:ss]	00:00:07.000	
Number of satellites used in the fix	10	14
Horizontal dilution of precision	0.7	1.0
Antenna height above mean sea level [m]	5.494	10.425
Speed over ground from successive positions [m s-1]	0.00	4.81
Horizontal acceleration [m s-2]	-0.455	0.519

Products

File	Records
SP2613_bestres.geoCSV	56925
SP2613_1min.geoCSV	475
SP2613_control.geoCSV	5

Quality control

0 of 56925 records flagged; a record can have more than one flag. Flagged records keep a leading # in bestres and are left out of 1min, control and the speeds.

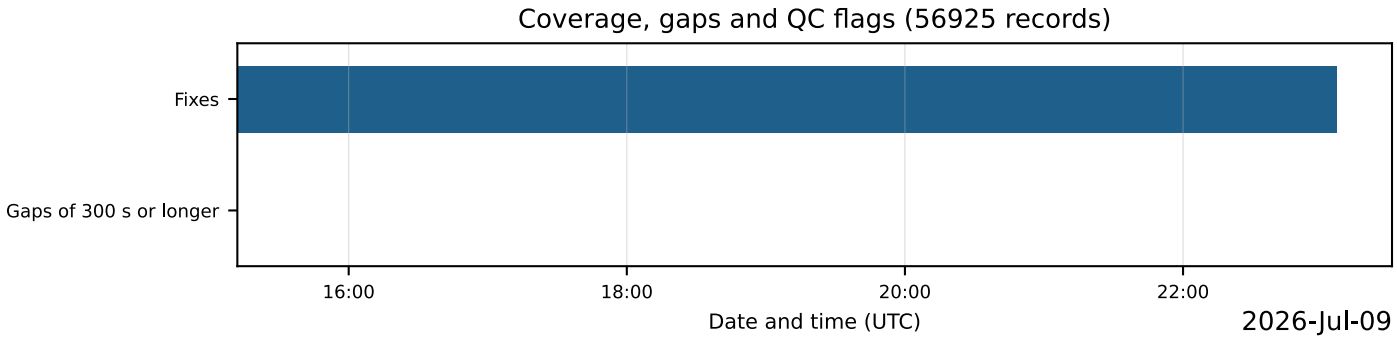
Reader and manifest errors

None reading the nav files or checking them against the manifest.

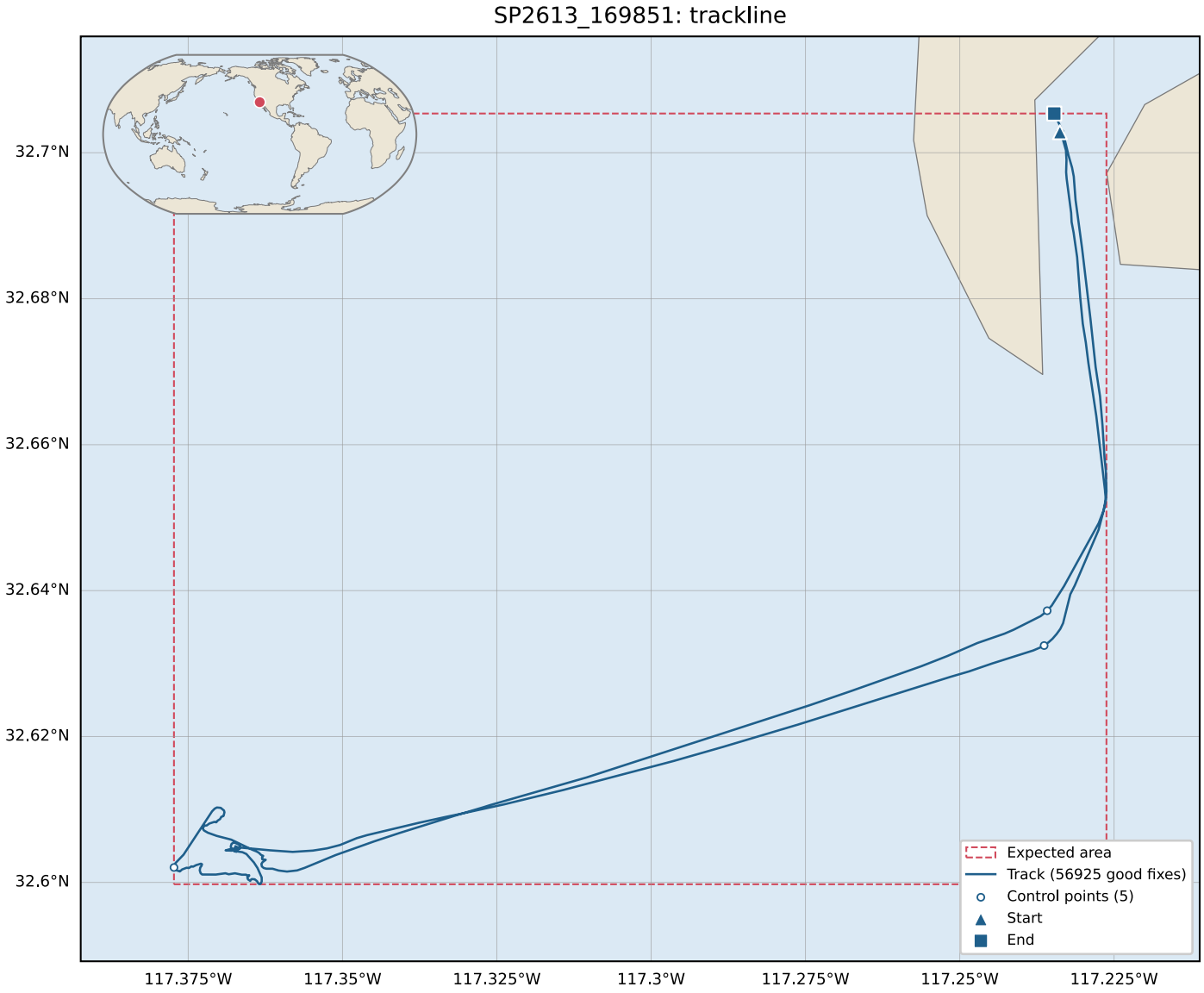
Gaps of 300 s (00:05:00) or longer

No gaps of 300 s (00:05:00) or longer.

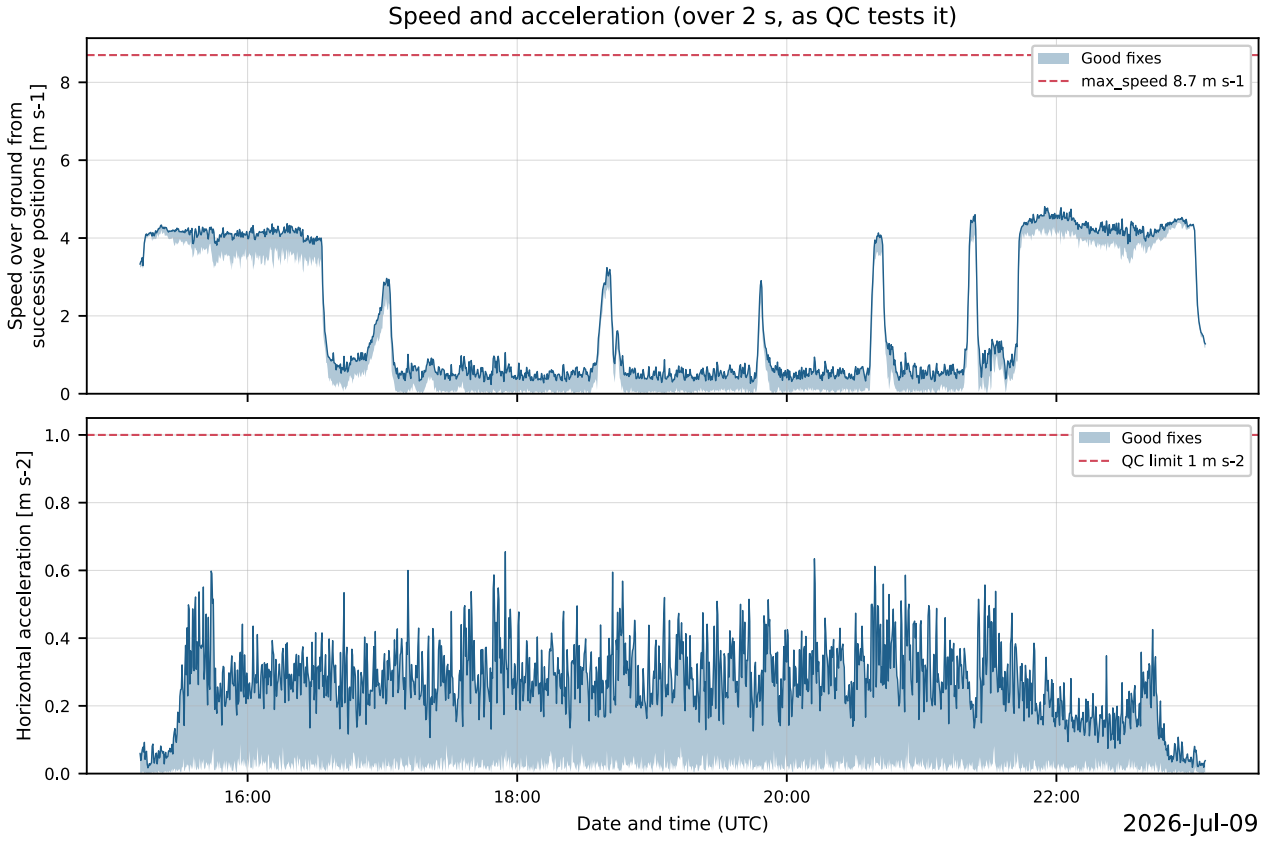
Coverage, gaps and flags



Trackline



Speed and acceleration



Sampling and fix quality

